

# Google Cloud Platform (GCP)

Cloud Computing Notes — Knowledge Base

---

## 1. Overview

Google Cloud Platform is Google's suite of cloud computing services, built on the same infrastructure Google uses internally for products like Search and YouTube. GCP is particularly well regarded for data analytics, machine learning, and Kubernetes (which originated at Google).

## 2. Core Services by Category

| Category       | Key Services                                    |
|----------------|-------------------------------------------------|
| Compute        | Compute Engine, Cloud Run, Cloud Functions, GKE |
| Storage        | Cloud Storage, Persistent Disk                  |
| Database       | Cloud SQL, Firestore, Bigtable, Spanner         |
| Data/Analytics | BigQuery, Dataflow, Pub/Sub                     |
| AI/ML          | Vertex AI, AutoML                               |

## 3. Distinctive Strengths

- BigQuery — serverless, highly scalable data warehouse for fast SQL analytics over massive datasets.
- GKE (Google Kubernetes Engine) — considered one of the most mature managed Kubernetes offerings, since Google created Kubernetes.
- Strong global network backbone, giving low-latency data transfer between regions.
- Spanner offers globally distributed, strongly consistent relational database capability — a fairly unique combination.

## 4. Core Concepts

- Projects — the base organizational unit for billing and resource management.
- IAM roles — fine-grained permission control similar to AWS IAM.
- Cloud Run — a popular serverless container platform that abstracts away cluster management.

## 5. Considerations

- Smaller market share and service breadth than AWS, though the core services are very strong.
- Fewer regions historically than AWS/Azure, though this gap has been closing.